

Project Roadmap (Implementation Levels)

LEVEL 1 — Basic Kubernetes Setup (Foundation)

 Goal: Just get cluster running

What you do:

Install:

Kubernetes

kubeadm

Create:

1 Master + 1 Worker node

Run:

Bash

kubectl get nodes

Output:

Cluster is working 

 Skills:

Linux + networking

Cluster setup

  Even this alone = mini DevOps project

LEVEL 2 — Run Containerized Job (Core K8s)

 Goal: Run any job inside Kubernetes

What you do:

Install:

Docker

Create a simple container (Python script)

Deploy using:

Kubernetes Jobs

Output:

Job runs → logs visible

 Skills:

Containers

Kubernetes Jobs

  Now it becomes a “cloud project”

LEVEL 3 — HPC Basics with MPI (CORE MVP)

 Goal: Parallel execution

What you do:

Install:

MPI (OpenMPI)

Run parallel program:

Multiple processes printing rank

Output:

Hello from process 0

Hello from process 1

...

🎯 Skills:

Parallel computing

HPC fundamentals

👉 🌟 THIS is your minimum impressive level

🔵 LEVEL 4 — Multi-Pod Parallel Execution (REAL HPC)

👉 Goal: Simulate distributed cluster

What you do:

Use Kubernetes Job with:

YAML

parallelism: 3

completions: 3

Run MPI across pods

Output:

Multiple pods executing simultaneously

🎯 Skills:

Distributed systems

Workload scaling

👉 🌟 Now you can say:

“Built HPC system on Kubernetes”

🟣 LEVEL 5 — Shared Storage + Coordination

👉 Goal: Real HPC behavior

What you do:

Add:

NFS

Store:

Input/output files shared across nodes

Output:

All nodes read/write same data

🎯 Skills:

Storage systems

Cluster coordination

👉 💡 This is where it becomes production-like

🔴 LEVEL 6 — Monitoring + Scaling (Production Grade)

👉 Goal: Make it industry-level

What you do:

Add:

Prometheus

Grafana

Enable:

Auto scaling (HPA)

Output:

Dashboard + metrics

Auto scaling cluster

🎯 Skills:

Observability

Performance tuning

👉 ⚡ This is top-tier project level

● LEVEL 7 (Optional FINAL BOSS)

👉 Only if you want to dominate competitions

Add:

Volcano Scheduler

AI-based scheduling

OSINT workload (your interest 🔥)

📊 Which Level Should YOU Target?

Based on your profile (CDAC + time constraints):

✅ Minimum safe:

👉 Level 3 (MPI working)

✅ Strong project:

👉 Level 4 (multi-pod HPC)

🔥 Standout:

👉 Level 5 (storage + real cluster behavior)

🏆 Elite:

👉 Level 6+